

CS4084

Quantum Computing

Week 08 – Lecture 16

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Outline

- 1 Quick Review: What We have Covered Last Lecture
- 2 Eigenvalues and Eigenvectors in Quantum Computing
- 3 Application: Quantum Phase Estimation (QPE)
- 4 Visualizing QPE
- 5 QPE Breakdown & Summary

Quick Review: What We Covered Last Lecture I

- **Quantum Circuits:** Graphical representation of quantum evolution.
 - **Wires** represent qubits; **Gates** represent unitary operations.
 - **Local vs Global:** Local gates act on single wires ($H \otimes I$); Global gates like CNOT operate across wires to enable **entanglement**.
- **Inner Products & Geometry:**
 - The inner product $\langle \phi | \psi \rangle$ defines the "overlap" between states.
 - **Orthogonality:** Two states are orthogonal if $\langle \phi | \psi \rangle = 0$.
 - **Orthonormality:** A basis $\{|v_i\rangle\}$ is orthonormal if $\langle v_i | v_j \rangle = \delta_{ij}$ (mutually orthogonal and unit length).
- **Projections:** Represented by $P = |\psi\rangle\langle\psi|$, used to project a state onto a specific subspace or basis vector.
- **Compound Systems & Tensor Products ($\otimes$):**
 - Joint states follow **lexicographical ordering**: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
 - **Multi-Qubit Space:** n qubits exist in $(\mathbb{C}^2)^{\otimes n}$.

Quick Review: What We Covered Last Lecture II

- **Matrix Tensor Product:** Joint operations use the **Kronecker product** ($M \otimes N$).
- **Bilinearity:** Tensor products are linear in every "slot," allowing scalars to move: $\alpha(|A\rangle \otimes |B\rangle) = (\alpha|A\rangle) \otimes |B\rangle$.
- **Probabilistic Operations:** Transformations on compound systems use **stochastic matrices** acting on the Cartesian product of state spaces.

The Eigenvalue Equation

In quantum mechanics, an **eigenvalue equation** describes a state that remains in the same "direction" after an operation, only changing by a scalar factor.

Mathematical Definition

For an operator (matrix) A and a non-zero vector $|\psi\rangle$:

$$A|\psi\rangle = \lambda|\psi\rangle$$

- $|\psi\rangle$ is the **eigenvector** (or eigenstate).
- λ is the **eigenvalue** (a complex scalar).

Physical Significance:

- **Measurement Outcomes:** The only possible results of measuring an observable A are its eigenvalues $\{\lambda_i\}$.
- **State Collapse:** After measurement, the system collapses into the corresponding eigenvector $|\psi_i\rangle$.

Example: The Pauli-Z Basis

The Z gate defines the standard computational basis $\{|0\rangle, |1\rangle\}$. Let's verify their eigenvalues:

For State $|0\rangle$:

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Eigenvalue $\lambda = +1$

For State $|1\rangle$:

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Eigenvalue $\lambda = -1$

- **Conclusion:** $|0\rangle$ and $|1\rangle$ are the eigenstates of the Z operator.
- When we measure in the Z -basis, we observe $+1$ (mapped to bit 0) or -1 (mapped to bit 1).

Example: The Pauli-X (NOT) Gate

What are the eigenstates of the X operator? Recall $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

The eigenstates are the "plus" and "minus" states:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Mathematical Verification:

$$X|+\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (+\mathbf{1})|+\rangle$$

$$X|-\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = (-\mathbf{1})|-\rangle$$

Takeaway: The same qubits can have different eigenvalues/eigenvectors depending on the **basis** (operator) we choose to measure.

Final Summary: Measurement & Phase

- **Projective Measurement:** To find the probability of an outcome λ_i , we project the current state $|\phi\rangle$ onto the eigenstate $|\psi_i\rangle$:

$$Pr(\lambda_i) = |\langle\psi_i|\phi\rangle|^2$$

- **Global Phase:** If $A|\psi\rangle = \lambda|\psi\rangle$, then $A(e^{i\theta}|\psi\rangle) = \lambda(e^{i\theta}|\psi\rangle)$. Because global phases cancel out in probability calculations, $e^{i\theta}|\psi\rangle$ is physically identical to $|\psi\rangle$.
- **No-Cloning Theorem:** Because unknown quantum states are superpositions of eigenstates, linear operators cannot "copy" them without destroying the original superposition.

Next Lecture: Diving into Quantum Entanglement and Bell States.

What is Quantum Phase Estimation?

The goal of QPE is to find the **eigenvalue** of a unitary operator U given its eigenstate $|\psi\rangle$.

The Mathematical Setup

Since U is unitary, its eigenvalues must have a magnitude of 1. We can write:

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$$

where θ is the **phase** we want to estimate (where $0 \leq \theta < 1$).

Why is this useful?

- **Chemistry:** If $U = e^{-iHt}$, finding θ gives us the energy levels (eigenvalues) of the Hamiltonian H .
- **Cryptography:** In Shor's algorithm, QPE is used to find the period of a function, which is encoded as a phase.

The QPE Circuit Architecture

QPE uses two registers of qubits:

- 1 **Register 1 (Counting):** n qubits initialized in $|0\rangle$. These will eventually hold the binary representation of the phase θ .
- 2 **Register 2 (State):** Initialized in the eigenstate $|\psi\rangle$.

Key Steps:

- **Superposition:** Apply Hadamard gates to Register 1.
- **Controlled-Unitaries:** Apply U^{2^j} operations controlled by Register 1. This "kicks" the phase information into the counting qubits (**Phase Kickback**).
- **Inverse QFT:** Apply the Inverse Quantum Fourier Transform ($QFT^\dagger$) to Register 1 to translate the phase from the "Fourier basis" back to the computational basis.

QPE Example: T -Gate Phase

Consider the T gate: $T|1\rangle = e^{i\pi/4}|1\rangle$. We want to find θ .

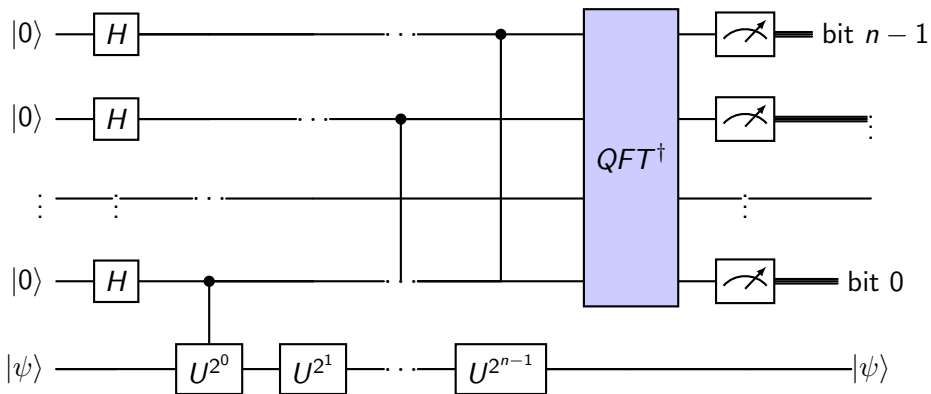
$$e^{2\pi i\theta} = e^{i\pi/4} \implies 2\pi\theta = \frac{\pi}{4} \implies \theta = \frac{1}{8}$$

Binary Representation: $\theta = \frac{1}{8}$ is written in binary as 0.001.

- With $n = 3$ counting qubits, the QPE circuit will output the state $|001\rangle$.
- Measuring this gives us 1, and we calculate $\theta = \frac{1}{2^3} = 0.125$.

QPE turns a global/relative phase into a measurable bitstring!

The QPE Circuit Architecture



Deep Dive: The QPE Architecture

The Quantum Phase Estimation algorithm relies on the precise interaction between two distinct registers:

- **Top Register (Counting):** Consists of n qubits initialized in $|0\rangle$. This register acts as the "storage" for the binary decimal representation of the phase θ .
- **Bottom Register (State):** Holds the eigenstate $|\psi\rangle$. Because $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, this register technically remains unchanged, but its presence allows the phase to be "kicked back."
- **Controlled- U Cascade:** The heart of the circuit. By applying U^{2^j} (successive doubling), we encode the phase into the counting qubits at different scales of precision.
- **$QFT^\dagger$ (Inverse Fourier Transform):** The final step. It transforms the state from the *Fourier basis* (where information is hidden in phases) back to the *Computational basis* (where it can be read as a bitstring).

Summary: Eigenvalues & QPE Takeaways I

- **Eigenstates are Fixed Points:** An eigenstate $|\psi\rangle$ of an operator A does not change its "direction" when A is applied; it is only scaled by the **eigenvalue** λ .
- **Measurement is Eigen-based:** In Quantum Computing, measuring an observable always yields one of its eigenvalues and collapses the qubit into the corresponding eigenstate.
- **Unitary Eigenvalues:** For any unitary gate U , the eigenvalues always take the form $e^{i\phi}$. These "phases" are where quantum information is often stored.
- **QPE is a Universal Tool:** Quantum Phase Estimation is the bridge that turns an abstract phase (eigenvalue) into a readable classical bitstring.
- **Algorithmic Impact:** QPE is the fundamental subroutine for:
 - **Shor's Algorithm:** Finding the period of a function.
 - **HHL Algorithm:** Solving linear systems of equations.
 - **Quantum Chemistry:** Calculating ground state energies of molecules.

Summary: Eigenvalues & QPE Takeaways II

- **Phase Kickback:** A critical concept where a controlled operation applies a phase to the *control* qubit, provided the *target* qubit is in an eigenstate.

Next Topic: Quantum Entanglement and Bell State Analysis